

In [1]:

```
import pandas
import numpy
```

In [4]:

```
arr2d = numpy.array([[1,2,3,4],[5,6,7,8]])
arr2d
```

Out[4]:

```
array([[1, 2, 3, 4],
       [5, 6, 7, 8]])
```

In [5]:

```
arr2d.shape
```

Out[5]:

```
(2, 4)
```

In [6]:

```
df = pandas.DataFrame([1,2,3,4,5])
print(df)
```

```
0
0 1
1 2
2 3
3 4
4 5
```

In [7]:

```
df1 = pandas.DataFrame([[1,2,3,4,5],[6,7,8,9,0]])
print(df1)
```

```
0 1 2 3 4
0 1 2 3 4 5
1 6 7 8 9 0
```

In [9]:

```
df2 = pandas.DataFrame([[1,2,3,4,5],[6,7,8,9,0]], index=['a','b'])
print(df2)
```

```
0 1 2 3 4
a 1 2 3 4 5
b 6 7 8 9 0
```

In [10]:

```
df3 = pandas.DataFrame([[1,2,3,4,5],[6,7,8,9,0]], index=['a','b'], columns=['a','b','c','d','e'])
print(df3)
```

```
   a  b  c  d  e
a  1  2  3  4  5
b  6  7  8  9  0
```

In [11]:

```
df3
```

Out[11]:

	a	b	c	d	e
a	1	2	3	4	5
b	6	7	8	9	0

Creating DataFrame using Dictionary

In [12]:

```
d = {
    'rollno':[1,2,3,4,5],
    'name':['Aman', 'Ajay', 'Arshnoor', 'Jatin', 'Vasu'],
    'class':['BCA', 'BCA', 'Btech', 'MCA', 'BCA']
}
df4 = pandas.DataFrame(d)
df4
```

Out[12]:

	rollno	name	class
0	1	Aman	BCA
1	2	Ajay	BCA
2	3	Arshnoor	Btech
3	4	Jatin	MCA
4	5	Vasu	BCA

In [13]:

```
print(df4)
```

```
   rollno  name  class
0        1  Aman   BCA
1        2  Ajay   BCA
2        3 Arshnoor Btech
3        4   Jatin   MCA
4        5   Vasu   BCA
```

CSV -> Comma Separated Values

In [15]:

```
df = pandas.read_csv("C:\\Users\\vmmin\\Desktop\\students.csv")
df
```

Out[15]:

	roll	name	class	marks
0	1	rahul	bca	56
1	1	vikas	bca	54
2	1	vasu	bca	76
3	1	ajay	bca	89
4	1	avinoor	bca	65
5	1	jatin	bca	90
6	1	vijay	bca	45
7	1	amit	bca	23
8	1	gaurav	bca	78

In [19]:

```
df1 = pandas.read_excel("C:\\Users\\vmmin\\Desktop\\students.xlsx")
df1
```

Out[19]:

	roll	name	class	marks
0	1	rahul	bca	56
1	1	vikas	bca	54
2	1	vasu	bca	76
3	1	ajay	bca	89
4	1	avinoor	bca	65
5	1	jatin	bca	90
6	1	vijay	bca	45
7	1	amit	bca	23
8	1	gaurav	bca	78

In [20]:

```
df['name']
```

Out[20]:

```
0    rahul
1    vikas
2     vasu
3     ajay
4  avinoor
5     jatin
6    vijay
7     amit
8   gaurav
Name: name, dtype: object
```

In [21]:

```
df.name
```

Out[21]:

```
0    rahul
1    vikas
2     vasu
3     ajay
4  avinoor
5     jatin
6    vijay
7     amit
8   gaurav
Name: name, dtype: object
```

In [22]:

```
df.to_csv('D://new_students.csv')
```

In [23]:

```
df.to_excel('D://new_students23.xlsx')
```

In [24]:

```
df.to_clipboard()
```

In [25]:

```
df.to_clipboard(sep=',')
```

In [28]:

```
df.to_dict()
```

Out[28]:

```
{'roll': {0: 1, 1: 1, 2: 1, 3: 1, 4: 1, 5: 1, 6: 1, 7: 1, 8: 1},
 'name': {0: 'rahul',
          1: 'vikas',
          2: 'vasu',
          3: 'ajay',
          4: 'avinoor',
          5: 'jatin',
          6: 'vijay',
          7: 'amit',
          8: 'gaurav'},
 'class': {0: 'bca',
            1: 'bca',
            2: 'bca',
            3: 'bca',
            4: 'bca',
            5: 'bca',
            6: 'bca',
            7: 'bca',
            8: 'bca'},
 'marks': {0: 56, 1: 54, 2: 76, 3: 89, 4: 65, 5: 90, 6: 45, 7: 23, 8: 78}}
```

In [30]:

```
df = pandas.read_csv('bollywood.csv')
df
```

Out[30]:

	Year	Title	Director	Cast	Genre	Release Month	Release Date
0	1920	Balika Badhu a.k.a. Baby Bride	P. T. Shaida (Tulsidutt Shaida)	shaida,pandit,ali,miss,tarak,bagcha,tulsidutt	social	NaN	NaN
1	1920	Daksha Yagna a.k.a. Sati Parvati	Vishnupant Divekar	prabha,khalil,vaidya	mythology	NaN	NaN
2	1920	Jalandhar Vrinda a.k.a. Sati Vrinda	G. V. Sane	unknown	mythology	NaN	NaN

In [31]:

```
df.head()
```

Out[31]:

	Year	Title	Director	Cast	Genre	Release Month
0	1920	Balika Badhu a.k.a. Baby Bride	P. T. Shaida (Tulsidutt Shaida)	shaida,pandit,ali,miss,tarak,bagcha,tulsidutt	social	Na
1	1920	Daksha Yagna a.k.a. Sati Parvati	Vishnupant Divekar	prabha,khalil,vaidya	mythology	Na
2	1920	Jalandhar Vrinda a.k.a. Sati Tuls Vrinda	G. V. Sane	unknown	mythology	Na
3	1920	Kansa Vadha a.k.a. End Of The Tyrant King	G. V. Sane	unknown	mythology	Na
4	1920	Katorabhar Khoon a.k.a.The Stinger Stung a.k.a...	S. N. Patankar	koregaonkar,gokhale,vyas,trymbakrao,pradhan,ta...	social	Na



In [32]:

```
df.head(3)
```

Out[32]:

	Year	Title	Director	Cast	Genre	Release Month	Release Date
0	1920	Balika Badhu a.k.a. Baby Bride	P. T. Shaida (Tulsidutt Shaida)	shaida,pandit,ali,miss,tarak,bagcha,tulsidutt	social	NaN	1920-01-01
1	1920	Daksha Yagna a.k.a. Sati Parvati	Vishnupant Divekar	prabha,khalil,vaidya	mythology	NaN	1920-01-01
2	1920	Jalandhar Vrinda a.k.a. Sati Tulsi Vrinda	G. V. Sane	unknown	mythology	NaN	1920-01-01



In [33]:

```
df.head(9)
```

Out[33]:

	Year	Title	Director	Cast	Genre	Rele Mo
0	1920	Balika Badhu a.k.a. Baby Bride	P. T. Shaيدا (Tulsidutt Shaيدا)	shaida,pandit,ali,miss,tarak,bagcha,tulsidutt	social	↑
1	1920	Daksha Yagna a.k.a. Sati Parvati	Vishnupant Divekar	prabha,khalil,vaidya	mythology	↑
2	1920	Jalandhar Vrinda a.k.a. Sati Tuls Vrinda	G. V. Sane	unknown	mythology	↑
3	1920	Kansa Vadha a.k.a. End Of The Tyrant King	G. V. Sane	unknown	mythology	↑
4	1920	Katorabhar Khoon a.k.a.The Stinger Stung a.k.a...	S. N. Patankar	koregaonkar,gokhale,vyas,trymbakrao,pradhan,ta...	social	↑
5	1920	Krishna Sudama a.k.a. Shri Krishna Sudama	Vishnupant Divekar	sakina,ali,moti,prabha,khalil,kesarbai,miya,va...	mythology	↑
6	1920	Lava Kusha	R. Nataraja Mudaliar	unknown	mythology	↑
7	1920	Mahabharata	Rustomji Dhotiwala	unknown	unknown	↑
8	1920	Mrichhakatik	Suchet Singh	kanjibhai,rathod	unknown	↑

In [34]:

```
df.tail()
```

Out[34]:

	Year	Title	Director	Cast	Genre	Release Month	Release Date	Highest Grosser By Year(in crores)
9677	2019	Panipat	Ashutosh Gowariker	Sanjay DuttArjun KapoorKriti SanonKabir Bedi	Action/battle	DEC	6.0	NaN
9678	2019	Pati, Patni Aur Woh	Mudassar Aziz	Kartik AaryanBhumi PednekarAnanya Panday	Comedy	DEC	6.0	NaN
9679	2019	Mardaani 2	Gopi Puthran	Rani MukherjeeVikram Singh ChauhanShruti Bapna...	Crime Thriller	DEC	13.0	NaN
9680	2019	Dabangg 3	Prabhu Deva	Salman KhanSonakshi SinhaArbaaz KhanMahie Gill...	Action	DEC	20.0	NaN
9681	2019	Good News	Raj Mehta	Akshay KumarKareena KapoorDiljit DosanjhKiara ...	Comedy	DEC	27.0	NaN

In [35]:

```
df.tail(8)
```

Out[35]:

	Year	Title	Director	Cast	Genre	Release Month	Release Date	Highest Grosser By Year(in crores)
9674	2019	Pagalpanti	Anees Bazmee	Anil KapoorJohn AbrahamIleana D'CruzArshad War...	Comedy	NOV	22.0	NaN
9675	2019	Dil Bechara	Mukesh Chhabra	Sushant Singh RajputSanjana SanghiSaif Ali Khan	Romantic drama	NOV	29.0	NaN
9676	2019	Jawaani Jaaneman	Nitin Kakkar	Saif Ali KhanTabuAlaia Furniturewalla	Romantic drama	NOV	29.0	NaN
9677	2019	Panipat	Ashutosh Gowariker	Sanjay DuttArjun KapoorKriti SanonKabir Bedi	Action/battle	DEC	6.0	NaN
9678	2019	Pati, Patni Aur Woh	Mudassar Aziz	Kartik AaryanBhumi PednekarAnanya Panday	Comedy	DEC	6.0	NaN
9679	2019	Mardaani 2	Gopi Puthran	Rani MukherjeeVikram Singh ChauhanShruti Bapna...	Crime Thriller	DEC	13.0	NaN
9680	2019	Dabangg 3	Prabhu Deva	Salman KhanSonakshi SinhaArbaaz KhanMahie Gill...	Action	DEC	20.0	NaN
9681	2019	Good News	Raj Mehta	Akshay KumarKareena KapoorDiljit DosanjhKiara ...	Comedy	DEC	27.0	NaN

In []:

